THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 2

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The number of covalent bonds in propane C_3H_8 is: 1

(a) 8 **(b)** 9 **(c)** 10 **(d)** 11

2. In $ZnO + C \longrightarrow Zn + CO$, the reducing agent is: 1

(a) C **(b)** ZnO **(c)** Zn **(d)** CO

3. $2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$ is a: 1

(a) displacement **(b)** thermal decomposition

(c) combination **(d)** neutralisation

4. Electronic configuration of an element X is 2, 8, 2. It forms an ion of charge: 1

(a) +1 **(b)** +2 **(c)** -2 **(d)** 0

5. CH_3CH_2OH with alkaline $KMnO_4$ and heat gives: 1

(a) $CH_2{=}CH_2$ **(b)** $CH_3COOC_2H_5$ **(c)** CH_3COOH **(d)** CH_3CH_3

6. Hydrogen and oxygen are obtained in the ratio 2 : 1 by volume because: 1

(a) water contains 2 hydrogen atoms for each oxygen atom

(b) oxygen is more soluble

(c) oxygen is heavier

(d) hydrogen is lighter

7. Which gas is evolved when $NaOH$ reacts with zinc? 1

(a) hydrogen **(b)** chlorine **(c)** oxygen **(d)** carbon dioxide

8. Zinc coating protects iron even if scratched because zinc: 1

(a) is heavy **(b)** is non-metallic

(c) is less reactive than iron **(d)** is more reactive than iron

9. Carbon cannot form C_4^+ because: 1

(a) it is a gas

(b) it has 8 valence electrons

(c) it has 2 valence electrons

(d) high energy is needed to remove 4 electrons

10. The coefficient of O_2 in the balanced equation $KClO_3 \longrightarrow KCl + O_2$ is: 1

(a) 2 **(b)** 1 **(c)** 3 **(d)** 4

11. Which of the following pairs of solutions will produce a neutral salt solution? 1

(a) HCl and NaOH **(b)** CH_3COOH and NaOH **(c)** HCl and NH_4OH **(d)** H_2CO_3 and NaOH

12. Which of the following reactions will take place? 1

(a) $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ **(b)** $Au + FeSO_4 \longrightarrow Au_2(SO_4)_3 + Fe$ **(c)** $Cu + ZnSO_4 \longrightarrow CuSO_4 + Zn$ **(d)** $Ag + CuSO_4 \longrightarrow Ag_2SO_4 + Cu$

13. The number of shared electron pairs in ethene (C_2H_4) is: 1

(a) 4 **(b)** 6 **(c)** 8 **(d)** 5

14. $2\,Pb(NO_3)_2 \longrightarrow 2\,PbO + 4\,NO_2 + O_2$ is a: 1

(a) displacement **(b)** photolysis

(c) precipitation **(d)** thermal decomposition

15. On passing excess CO_2 in lime water, the milkiness: 1

(a) disappears **(b)** remains same **(c)** turns pink **(d)** increases

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. How does a farmer treat acidic soil? Give reason. 2

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17. Why does an ionic compound conduct electricity in aqueous solution but not in the solid state? 2

18. How will you test for unsaturation in a hydrocarbon? 2

19. Why are silver bromide and silver chloride kept in dark bottles? 2

20. Why is baking powder used in making cakes soft and spongy? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Compare the properties of soap and detergent. 3

22. In the electrolysis of water, name the gas at each electrode, the ratio of volumes and the equation. 3

23. Why does vinegar remove limescale (calcium carbonate)? Write the reaction. 3

24. Explain two reasons why iron is galvanised. 3

OR

What is a covalent bond? Show formation of Cl_2 and O_2. 3

25. Balance $NH_3 + O_2 \longrightarrow NO + H_2O$ and mention the type of reaction. 3

OR

Explain how a solution of pH 7 can be made from pH 1 and pH 13 solutions, giving the principle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain thermal, electrolytic and photolytic decomposition with one equation each. State which of the three is used in photography. 5

OR

Explain the preparation and setting of plaster of Paris with equations. State its uses. 5

27. Explain how corrosion is prevented in 5
(a) iron bridges
(b) aluminium utensils
(c) ships, with reasons.

OR

Describe the substitution and addition reactions and the conditions with equations; also classify the reaction of ethene with bromine. 5

END OF QUESTION PAPER — Total: 50 Marks

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